Nassau Candy Distributor

Executive Summary for Government Stakeholders

Factory-to-Customer Route Efficiency Analysis

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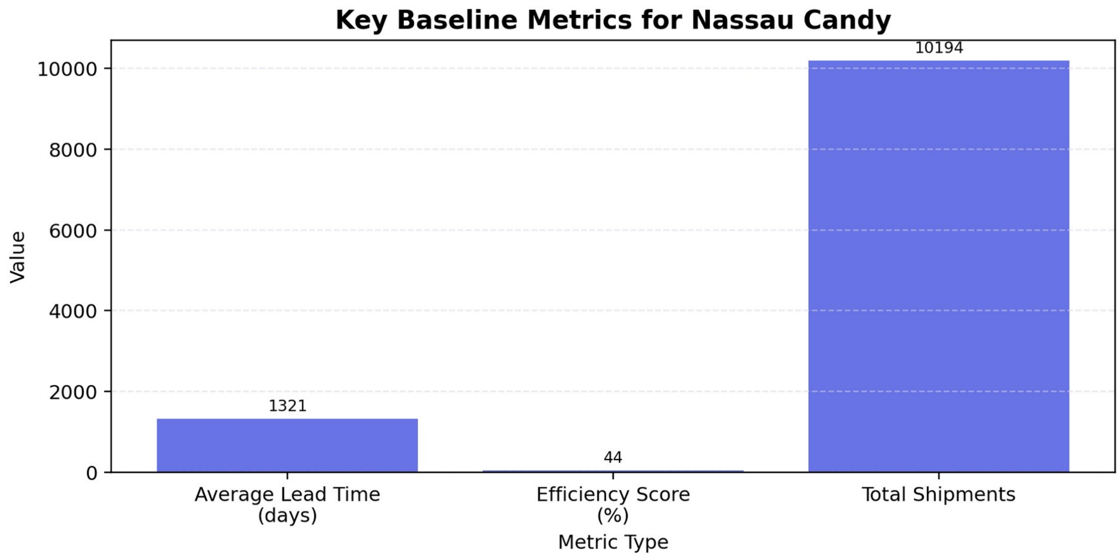
1. Purpose

This executive summary presents the key findings from the Nassau Candy Distributor route efficiency analysis. The project evaluates recorded lead time between customer order date and ship date, compares route and shipping-mode performance, and identifies operational bottleneck candidates that require further review.

Data note: Shipping Lead Time is calculated as Ship Date minus Order Date using the dates in the supplied dataset. The source data contains Ship Date values from 2026 to 2030 while Order Date values run from 2024 to 2025. Because actual customer delivery dates and carrier scan events are not included, this report treats the metric as recorded order-to-shipment lead time and identifies bottleneck candidates rather than proving infrastructure causes.

2. Key Findings

- The dataset contains 10,194 shipment records across 59 states/provinces and 4 business regions.
- The average recorded lead time is 1,320.8 days and the median is 1,274 days.
- Using the system average of 1,321 days as the delay threshold, 33.1% of shipments are above the threshold.
- First Class and Same Day show higher average recorded lead time than Standard Class, which indicates a need to review order processing and priority handling before shipment release.
- The highest total bottleneck impact is concentrated in high-volume routes such as Lot's O' Nuts -> California, Wicked Choccy's -> California, and Lot's O' Nuts -> New York.



3. Route and Geographic Performance Summary

The table below focuses on routes with strong operational relevance. High-volume bottleneck routes are not always the slowest on average, but they create the largest total network impact because they carry many shipments.

Route	Shipments	Avg Lead Time	Bottleneck Score
Lot's O' Nuts -> California	1,125	1,316.9	1,481,524.0
Wicked Choccy's -> California	823	1,320.3	1,086,629.0
Lot's O' Nuts -> New York	645	1,330.5	858,183.0
Lot's O' Nuts -> Texas	569	1,310.5	745,652.0
Wicked Choccy's -> New York	436	1,311.9	571,981.0
Wicked Choccy's -> Texas	388	1,315.6	510,456.0

At a state/province level, high-volume areas with above-average recorded lead time should be investigated before low-volume outliers. This keeps improvement efforts focused on routes that are more likely to affect customer experience and operational cost.

State/Province	Shipments	Avg Lead Time	Median Lead Time
Tennessee	183	1,391.5	1,276.0
Indiana	149	1,381.5	1,274.0
Washington	506	1,360.7	1,275.0
Connecticut	82	1,357.5	1,274.0
Maryland	105	1,356.8	1,274.0
Wisconsin	110	1,343.0	1,274.0

4. Stakeholder Implications

The findings indicate that logistics performance should be reviewed at route level rather than only at national or regional level. Some routes carry high volume and therefore create greater system-wide impact even when their average lead time is close to the network average.

For government or infrastructure stakeholders, this analysis should be treated as an early-warning view of logistics pressure points. It does not prove that roads, ports, or public infrastructure are the direct cause of delays. Further evidence would require carrier transit scans, warehouse dwell-time data, delivery confirmation dates, weather records, and lane-level transportation data.

5. Strategic Recommendations

- Deploy the Streamlit dashboard for recurring review of route lead time, delay percentage, and bottleneck score.
- Prioritize investigation of high-volume routes with high total bottleneck impact, especially California, New York, Texas, Pennsylvania, and Washington routes from the largest factories.
- Audit internal processing for First Class and Same Day orders to confirm whether priority shipments are being released faster than standard orders.
- Create a monthly route performance review process using a consistent threshold such as system average lead time or the 75th percentile.
- Collect additional delivery and carrier data before making infrastructure investment claims or public-policy conclusions.

6. Conclusion

The analysis gives Nassau Candy Distributor a clearer view of route-level performance and identifies where operational review should begin. The strongest improvement opportunity is to monitor high-volume routes, investigate priority shipping workflows, and use the dashboard as a repeatable decision-support tool. With additional delivery and carrier data, future analysis can separate internal processing delays from external transportation constraints.